Effect of dietary soy intake on breast cancer risk according to menopause and hormone receptor status.

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BACKGROUND: Although high soy consumption may be associated with lower breast cancer risk in Asian populations, findings from epidemiological studies have been inconsistent. OBJECTIVE: We investigated the effects of soy intake on breast cancer risk among Korean women according to their menopausal and hormone receptor status. METHODS: We conducted a case-control study with 358 incident breast cancer patients and 360 age-matched controls with no history of malignant neoplasm. Dietary consumption of soy products was examined using a 103-item food frequency questionnaire. RESULTS: The estimated mean intakes of total soy and isoflavones from this study population were 76.5 g per day and 15.0 mg per day, respectively. Using a multivariate logistic regression model, we found a significant inverse association between soy intake and breast cancer risk, with a dose-response relationship (odds ratios (OR) (95% confidence interval (CI)) for the highest vs the lowest intake quartile: 0.36 (0.20-0.64)). When the data were stratified by menopausal status, the protective effect was observed only among postmenopausal women (OR (95% CI) for the highest vs the lowest intake quartile: 0.08 (0.03-0.22)). The association between soy and breast cancer risk did not differ according to estrogen receptor (ER)/progesterone receptor (PR) status, but the estimated intake of soy isoflavones showed an inverse association only among postmenopausal women with ER+/PR+ tumors. CONCLUSIONS: Our findings suggest that high consumption of soy might be related to lower risk of breast cancer and that the effect of soy intake could vary depending on several factors.